

Patients undergoing chemotherapy frequently experience a number of side effects in response to the cytotoxic drugs administered during treatment, including nausea and/or vomiting (N/V), hair loss, fatigue, cognitive dysfunction, anemia, and others. However, there is ample evidence that learning and cognition may play a role in the development of some of these side effects. For example, 20%–30% of patients undergoing chemotherapy continue to experience N/V merely in the setting and presence of chemotherapy equipment even after taking antiemetic medication.

Three studies were conducted to better understand the acquisition of a learned, anticipatory nausea (AN) response. Studies 1 and 2 used the classical conditioning principle of *overshadowing*, in which a powerful stimulus is presented alongside a less salient stimulus during the acquisition phase. This results in a stronger relationship between the salient stimulus and the response, as compared with the less salient stimulus and response.

In Study 1, nausea was induced in 50 healthy volunteers through motion sickness in a rotating chair. Subjects were divided into two groups and drank specific beverages prior to the rotation: One group received a different novel-tasting beverage (eg, elderberry, sloe) each day, and the other group drank

only water. Nausea-related symptoms were recorded before and after the rotation. Results showed that subjects who drank the novel-tasting beverages reported less AN than those who drank water.

Study 2 tested the protocol from Study 1 in cancer patients undergoing chemotherapy. Immediately prior to chemotherapy infusion, one group received a different novel-tasting beverage each day and the other group drank only water. Results again demonstrated that subjects in the group with different novel-tasting beverages experienced less AN than those who drank water.

Study 3 researchers investigated whether patients' pretreatment expectations of developing nausea or

medication-induced post-chemotherapy nausea (PN) were a better predictor of subsequent AN. Before their first round of chemotherapy (session #1), subjects were surveyed about their degree of certainty of developing nausea. PN was measured after session #1, and then AN was measured before session #3 (Figure 1).

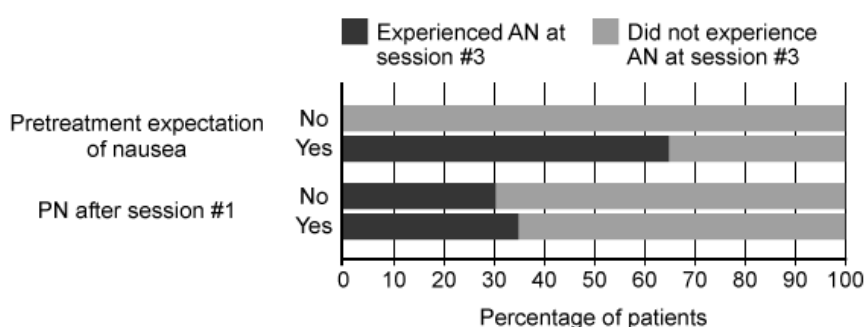


Figure 1 AN before session #3

Based on the passage, the control subjects in Study 2 could also develop which additional conditioned response?

- ☐ A. Hair loss from the chemotherapy medication [4%]
- ✓ ☒ B. Fatigue when entering the chemotherapy room [61%]
- ☐ C. Difficulty concentrating when drinking novel-flavored beverages [13%]
- ☐ D. Nausea and/or vomiting while riding a rotating ride at an amusement park [19%]

Correct

62% answered correctly

Explanation:

Associative learning: classical conditioning

Before acquisition (response is innate)

Unconditioned stimulus (UCS) → Unconditioned response (UCR)
eg, cytotoxic medication → eg, N/V, fatigue

During acquisition

Unconditioned stimulus + Neutral stimulus (NS) → Unconditioned response (UCR)
eg, cytotoxic medication + eg, sight of chemotherapy equipment → eg, N/V, fatigue

After acquisition (response is learned)

Conditioned stimulus (CS) → Conditioned response (CR)
eg, sight of chemotherapy equipment → eg, N/V, fatigue

In **classical conditioning**, a previously neutral stimulus causes an organism to react in a manner similar to the reaction to a natural or unconditioned stimulus after the two stimuli have been paired. Unconditioned stimuli (eg, meat) are biologically arousing, causing an innate (untrained) reaction called an **unconditioned response** (eg, salivation). Neutral stimuli (eg, bell) do not produce a meaningful response. After being paired with the unconditioned stimulus, the previously neutral stimulus, now termed the conditioned stimulus, causes the organism to react with a learned, or **conditioned response** (eg, salivation in response to a bell). The conditioned response is typically similar to the unconditioned response.

For patients receiving chemotherapy, the room, apparatus, smells, and other characteristics of the context (neutral stimuli) could become associated with the experience of nausea because it is in this environment that the cytotoxic medication (unconditioned stimulus) caused the nausea (unconditioned response). After learning, those contextual characteristics of the room (conditioned stimulus) will provoke AN (conditioned response) before the cytotoxic medication is administered. Similarly, a patient could develop the additional conditioned response of feeling fatigue when entering the chemotherapy room.

(Choices A and D) Unconditioned responses occur naturally (or biologically) as a result of unconditioned stimuli; they are not learned. Hair loss follows chemotherapy medication because it destroys the hair follicle cells, and nausea naturally results from motion sickness.

(Choice C) Control subjects in Study 2 received only water, so they would not develop a conditioned response (ie, difficulty concentrating) in association with novel-flavored beverages.

Educational objective:

Unconditioned responses occur naturally, without learning (eg, nausea in response to cytotoxic medication). Conditioned responses occur when a neutral stimulus (eg, chemotherapy equipment) is paired with an unconditioned stimulus (eg, cytotoxic medication).

Time spent:0 sec

Subject:

Behavioral Sciences

QID:401401

System:

Learning, Memory, and Cognition

Patients undergoing chemotherapy frequently experience a number of side effects in response to the cytotoxic drugs administered during treatment, including nausea and/or vomiting (N/V), hair loss, fatigue, cognitive dysfunction, anemia, and others. However, there is ample evidence that learning and cognition may play a role in the development of some of these side effects. For example, 20%–30% of patients undergoing chemotherapy continue to experience N/V merely in the setting and presence of chemotherapy equipment even after taking antiemetic medication.

Three studies were conducted to better understand the acquisition of a learned, anticipatory nausea (AN) response. Studies 1 and 2 used the classical conditioning principle of *overshadowing*, in which a powerful stimulus is presented alongside a less salient stimulus during the acquisition phase. This results in a stronger relationship between the salient stimulus and the response, as compared with the less salient stimulus and response.

In Study 1, nausea was induced in 50 healthy volunteers through motion sickness in a rotating chair. Subjects were divided into two groups and drank specific beverages prior to the rotation: One group received a different novel-tasting beverage (eg, elderberry, sloe) each day, and the other group drank only water. Nausea-related symptoms were recorded before and after the rotation. Results showed that subjects who drank the novel-tasting beverages reported less AN than those who drank water.

Study 2 tested the protocol from Study 1 in cancer patients undergoing chemotherapy. Immediately prior to chemotherapy infusion, one group received a different novel-tasting beverage each day and the other group drank only water. Results again demonstrated that subjects in the group with different novel-tasting beverages experienced less AN than those who drank water.

Study 3 researchers investigated whether patients' pretreatment

expectations of developing nausea or medication-induced post-chemotherapy nausea (PN) were a better predictor of subsequent AN. Before their first round of chemotherapy (session #1), subjects were surveyed about their degree of certainty of developing nausea. PN was measured after session #1, and then AN was measured before session #3 (Figure 1).

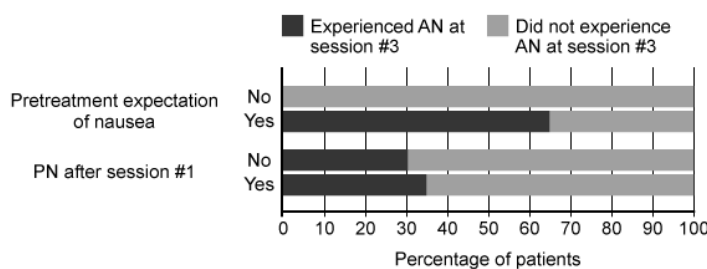


Figure 1 AN before session #3

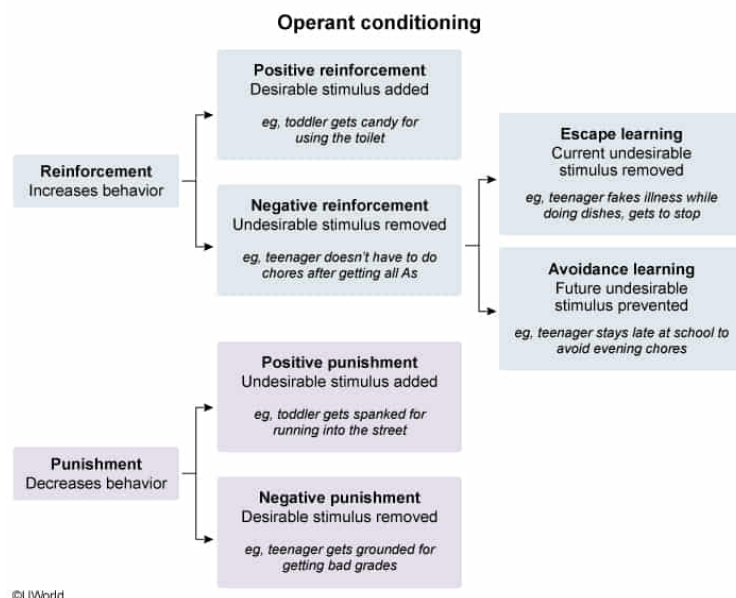
Which of the following is NOT accurate regarding antiemetic medications, which are administered to patients undergoing chemotherapy to treat or prevent N/V?

- ✓ ☒ A. An antiemetic that prevents a recurrence of N/V acts as positive reinforcement. [43%]
- ☐ B. An antiemetic that treats current PN acts as negative reinforcement. [32%]
- ☐ C. An antiemetic that treats current PN may produce escape learning. [10%]
- ☐ D. An antiemetic that prevents a recurrence of N/V may produce avoidance learning. [13%]

Correct

42% answered correctly

Explanation:



N/V is an aversive stimulus. Antiemetic medications can be administered while patients are experiencing N/V to treat their N/V, or before chemotherapy to prevent N/V entirely. Either scenario may result in **operant conditioning**, in which the likelihood of an individual repeating a behavior is influenced by the outcome of that behavior (ie, by reward or punishment).

Behavior is increased or maintained by reinforcement; **positive reinforcement** occurs when a *desirable* stimulus is applied, encouraging a behavior to happen again. For example, a young child receives a lollipop (desirable stimulus) for good behavior during a medical appointment, which increases the likelihood of future good behavior.

In terms of the answer choice that is **not** accurate regarding antiemetic medications, an antiemetic that prevents a recurrence of N/V is *inhibiting* an *undesirable* stimulus, not *adding* a *desirable* stimulus; therefore, it is **not** an example of positive reinforcement.

(Choice B) Negative reinforcement occurs when an *undesirable stimulus* is *withdrawn*, which encourages a behavior to happen again. For example, taking an antiemetic to treat (remove) current PN (unpleasant stimulus) increases the likelihood of the individual taking the medication in the future, which is negative reinforcement.

(Choice C) Escape learning occurs when an individual learns how to *terminate a current unpleasant stimulus*. For example, an antiemetic that treats current PN may produce escape learning, as the individual learns that taking the medication stops N/V (unpleasant stimulus).

(Choice D) Avoidance learning occurs when an individual figures out how to *prevent* experiencing *the unpleasant stimulus* in the *future*. For example, taking an antiemetic before chemotherapy, thereby preventing a recurrence of N/V (unpleasant stimulus), is an example of avoidance learning.

Educational objective:

In operant conditioning, the outcome of a behavior impacts the likelihood that the behavior will be repeated; behavior increases due to reinforcement, and it decreases due to punishment. Positive reinforcement refers to a desirable stimulus being applied, whereas negative reinforcement is an undesirable stimulus being withdrawn.

Time spent:0 sec

Subject:

Behavioral Sciences

QID:401402

System:

Learning, Memory, and Cognition

Patients undergoing chemotherapy frequently experience a number of side effects in response to the cytotoxic drugs administered during treatment, including nausea and/or vomiting (N/V), hair loss, fatigue, cognitive dysfunction, anemia, and others. However, there is ample evidence that learning and cognition may play a role in the development of some of these side effects. For example, 20%–30% of patients undergoing chemotherapy continue to experience N/V merely in the setting and presence of chemotherapy equipment even after taking antiemetic medication.

Three studies were conducted to better understand the acquisition of a learned, anticipatory nausea (AN) response. Studies 1 and 2 used the classical conditioning principle of *overshadowing*, in which a powerful stimulus is presented alongside a less salient stimulus during the acquisition phase. This results in a stronger relationship between the salient stimulus and the response, as compared with the less salient stimulus and response.

In Study 1, nausea was induced in 50 healthy volunteers through motion sickness in a rotating chair. Subjects were divided into two groups and drank specific beverages prior to the rotation: One group received a different novel-tasting beverage (eg, elderberry, sloe) each day, and the other group drank

only water. Nausea-related symptoms were recorded before and after the rotation. Results showed that subjects who drank the novel-tasting beverages reported less AN than those who drank water.

Study 2 tested the protocol from Study 1 in cancer patients undergoing chemotherapy. Immediately prior to chemotherapy infusion, one group received a different novel-tasting beverage each day and the other group drank only water. Results again demonstrated that subjects in the group with different novel-tasting beverages experienced less AN than those who drank water.

Study 3 researchers investigated whether patients' pretreatment expectations of developing nausea or

medication-induced post-chemotherapy nausea (PN) were a better predictor of subsequent AN. Before their first round of chemotherapy (session #1), subjects were surveyed about their degree of certainty of developing nausea. PN was measured after session #1, and then AN was measured before session #3 (Figure 1).

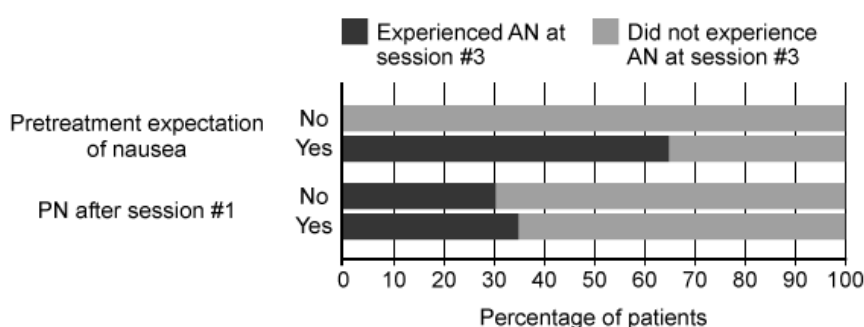


Figure 1 AN before session #3

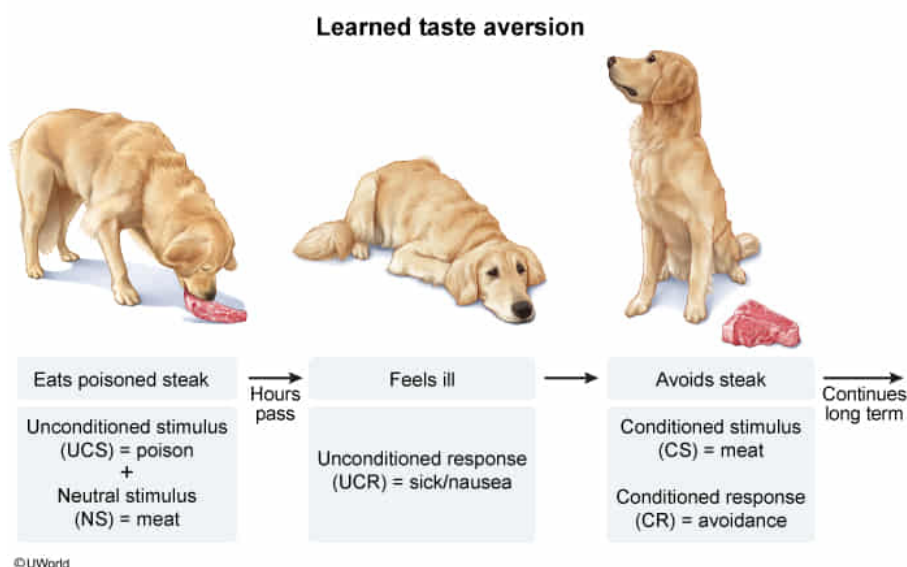
Based on the passage and the principles of taste aversion, which strategy should physicians suggest to their patients undergoing chemotherapy? On the day of chemotherapy, patients should:

- ☐ A. pair their favorite foods with a novel beverage. [31%]
- ☐ B. eat favorite foods several hours before chemotherapy treatment. [9%]
- ☐ C. only consume favorite beverages, not water. [11%]
- ✓ ☒ D. avoid eating their favorite foods entirely. [46%]

Correct

46% answered correctly

Explanation:



A conditioned **taste aversion** is a specific and powerful type of **classical conditioning** that occurs when an organism becomes ill after consuming something (eg, food, alcohol). Whatever was consumed prior to the N/V can become associated with the illness (even if it did not *cause* the illness) and is avoided afterward, long term. Taste aversions almost always link illness with foods (or smells), which is thought to be an evolutionary adaptation.

Of the given choices, a physician should suggest that patients avoid eating their favorite foods on the day of chemotherapy. This strategy will prevent patients from developing a taste aversion to these foods. If a favored food is consumed prior to the experience of nausea (resulting from chemotherapy), it is possible that the food will be permanently associated with the nausea.

(Choice A) Although there is a chance that a salient, novel-flavored beverage could overshadow the favorite food and become linked to the nausea, there is nothing in the passage that suggests it would *prevent* a taste aversion if paired with a favorite food.

(Choice B) Taste aversions can develop even with hours passing between food consumption and becoming ill. Consuming a favorite food hours before chemotherapy would not prevent the development of a taste aversion.

(Choice C) Patients could develop a taste aversion to their favorite beverages if consumed prior to becoming ill.

Educational objective:

Taste aversion is a specific and powerful type of classical conditioning that occurs after just one instance of becoming ill following food/beverage consumption.

Taste aversions are long lasting and can develop despite many hours passing between consumption and illness.

Time spent:0 sec

Subject:

Behavioral Sciences

QID:401403

System:

Learning, Memory, and Cognition

Patients undergoing chemotherapy frequently experience a number of side effects in response to the cytotoxic drugs administered during treatment, including nausea and/or vomiting (N/V), hair loss, fatigue, cognitive dysfunction, anemia, and others. However, there is ample evidence that learning and cognition may play a role in the development of some of these side effects. For example, 20%–30% of patients undergoing chemotherapy continue to experience N/V merely in the setting and presence of chemotherapy equipment even after taking antiemetic medication.

Three studies were conducted to better understand the acquisition of a learned, anticipatory nausea (AN) response. Studies 1 and 2 used the classical conditioning principle of *overshadowing*, in which a powerful stimulus is presented alongside a less salient stimulus during the acquisition phase. This results in a stronger relationship between the salient stimulus and the response, as compared with the less salient stimulus and response.

In Study 1, nausea was induced in 50 healthy volunteers through motion sickness in a rotating chair. Subjects were divided into two groups and drank specific beverages prior to the rotation: One group received a different novel-tasting beverage (eg, elderberry, sloe) each day, and the other group drank only water. Nausea-related symptoms were recorded before and after the rotation. Results showed that subjects who drank the novel-tasting beverages reported less AN than those who drank water.

Study 2 tested the protocol from Study 1 in cancer patients undergoing chemotherapy. Immediately prior to chemotherapy infusion, one group received a different novel-tasting beverage each day and the other group drank only water. Results again demonstrated that subjects in the group with different novel-tasting beverages experienced less AN than those who drank water.

Study 3 researchers investigated whether patients' pretreatment expectations of developing nausea or medication-induced post-chemotherapy nausea (PN) were a better predictor of subsequent AN. Before their first round of chemotherapy (session #1), subjects were surveyed about their degree of certainty of developing nausea. PN was measured after session #1, and then AN was measured before session #3 (Figure 1).

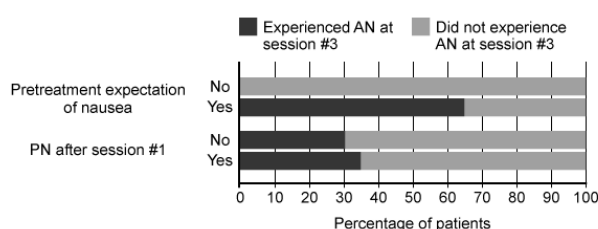


Figure 1 AN before session #3

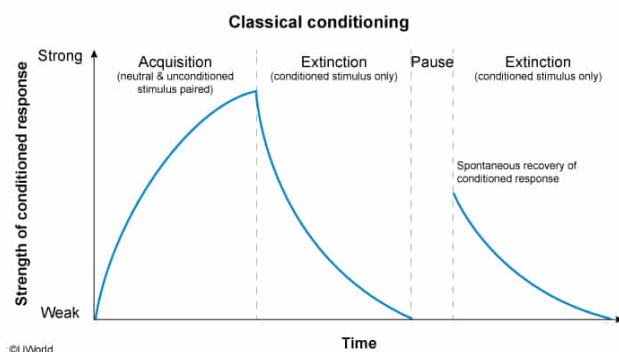
In a follow-up to Study 2, the control participants returned to the chemotherapy room to fill out surveys every day for a week after their final chemotherapy treatment. By the end of the week, participants' AN had significantly decreased. This illustrates which phenomenon?

- ☐ A. Acquisition [13%]
- ☐ B. Stimulus generalization [14%]
- ✓ ☒ C. Extinction [67%]
- ☐ D. Spontaneous recovery [5%]

Correct

67% answered correctly

Explanation:



In classical conditioning, the strength of the conditioned response tends to follow a pattern over time.

Extinction occurs when a conditioned response gradually stops occurring in the absence of the unconditioned stimulus. For example, a dog that has been conditioned to salivate (conditioned response) in response to the sound of a bell (conditioned stimulus) will gradually stop salivating to the sound of the bell if that sound is repeatedly presented without food.

If the control participants repeatedly return to the chemotherapy room to fill out surveys, they would experience the conditioned stimulus (ie, the room, equipment) in the absence of the unconditioned stimulus (ie, chemotherapy medication), which should result in a reduction of the conditioned response (ie, AN). This is an example of *extinction*.

(Choice A) Acquisition refers to the learning that takes place as an association is

formed between the unconditioned stimulus (eg, food) and the neutral stimulus (eg, bell).

(Choice B) Stimulus generalization occurs when the conditioned response is elicited by stimuli similar to the original conditioned stimulus. For example, a dog that has been conditioned to salivate in response to a bell may also salivate to a phone alert.

(Choice D) Spontaneous recovery occurs when an extinct response reappears after a period of time. For example, a dog whose salivating response has become extinct will salivate again on hearing a bell a few days after extinction has occurred.

Educational objective:

In classical conditioning, extinction occurs when the conditioned stimulus no longer produces the conditioned response in the absence of the unconditioned stimulus.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401404

System:
Learning, Memory, and Cognition

Patients undergoing chemotherapy frequently experience a number of side effects in response to the cytotoxic drugs administered during treatment, including nausea and/or vomiting (N/V), hair loss, fatigue, cognitive dysfunction, anemia, and others. However, there is ample evidence that learning and cognition may play a role in the development of some of these side effects. For example, 20%–30% of patients undergoing chemotherapy continue to experience N/V merely in the setting and presence of chemotherapy equipment even after taking antiemetic medication.

Three studies were conducted to better understand the acquisition of a learned, anticipatory nausea (AN) response. Studies 1 and 2 used the classical conditioning principle of *overshadowing*, in which a powerful stimulus is presented alongside a less salient stimulus during the acquisition phase. This results in a stronger relationship between the salient stimulus and the response, as compared with the less salient stimulus and response.

In Study 1, nausea was induced in 50 healthy volunteers through motion sickness in a rotating chair. Subjects were divided into two groups and drank specific beverages prior to the rotation: One group received a different novel-tasting beverage (eg, elderberry, sloe) each day, and the other group drank only water. Nausea-related symptoms were recorded before and after the rotation. Results showed that subjects who drank the novel-tasting beverages reported less AN than those who drank water.

Study 2 tested the protocol from Study 1 in cancer patients undergoing chemotherapy. Immediately prior to chemotherapy infusion, one group received a different novel-tasting beverage each day and the other group drank only water. Results again demonstrated that subjects in the group with different novel-tasting beverages experienced less AN than those who drank water.

Study 3 researchers investigated whether patients' pretreatment expectations of developing nausea or medication-induced post-chemotherapy nausea (PN) were a better predictor of subsequent AN. Before their first round of chemotherapy (session #1), subjects were surveyed about their degree of certainty of developing nausea. PN was measured after session #1, and then AN was measured before session #3 (Figure 1).

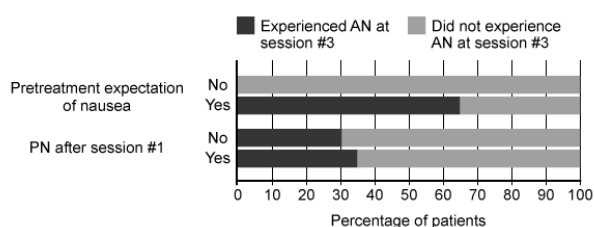


Figure 1 AN before session #3

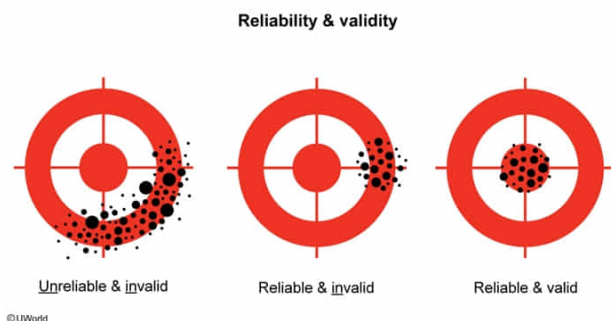
If the survey from Study 3 that assessed the patients' level of AN was determined to be reliable, which of the following statements is true?

- ☐ A. Researchers can confidently extrapolate from Study 3 to all patients with cancer. [5%]
- ☐ B. Survey administration is consistent across medical settings. [10%]
- ☐ C. The survey is an accurate measure of nausea symptoms. [18%]
- ✓ ☒ D. Nausea scores on the survey will be similar from one survey administration to the next. [65%]

Correct

64% answered correctly

Explanation:



A variety of criteria are used to judge the extent to which scientific findings are credible (ie, can be trusted or believed). Both the overall study design and particular measures used must be examined for reliability and validity.

Reliability refers to the **consistency** of an experiment or measure. Reliable measures produce **similar results** every time. For example, two MCAT examinations taken at the same time by the same person on two consecutive days should produce comparable scores. Similarly, if a study is well designed, it should produce similar results when repeated with new subjects. This is an indication that an experimental design is highly reliable, also referred to as **replicability** or reproducibility.

Validity refers to the **accuracy** of a study or measure (**Choice C**). Reliability is typically a necessary, but not sufficient, condition for validity; simply producing the same results does not mean your results reflect reality. For example, a broken digital thermometer might consistently read 2 degrees above the actual temperature (ie, it is reliable but not valid).

(Choice A) External validity, also known as generalizability, is the extent to which study results can be *applied outside the laboratory* to real-life situations. Experiments in which all variables are tightly controlled may have a high degree of reliability and internal validity but not be very generalizable to the real world.

(Choice B) Standardization refers to test administration and/or scoring being done in a uniform manner. Consistency of test administration and scoring allows researchers to better compare one subject or group to the next.

Educational objective:

Reliability describes the degree to which a measure or experiment is consistent (eg, a test produces a similar score for one person who takes it twice). Validity refers to the accuracy of a measure or experiment (eg, a test accurately assesses what it was designed to measure).

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401405

System:
Learning, Memory, and Cognition

Patients undergoing chemotherapy frequently experience a number of side effects in response to the cytotoxic drugs administered during treatment, including nausea and/or vomiting (N/V), hair loss, fatigue, cognitive dysfunction, anemia, and others. However, there is ample evidence that learning and cognition may play a role in the development of some of these side effects. For example, 20%–30% of patients undergoing chemotherapy continue to experience N/V merely in the setting and presence of chemotherapy equipment even after taking antiemetic medication.

Three studies were conducted to better understand the acquisition of a learned, anticipatory nausea (AN) response. Studies 1 and 2 used the classical conditioning principle of *overshadowing*, in which a powerful stimulus is presented alongside a less salient stimulus during the acquisition phase. This results in a stronger relationship between the salient stimulus and the response, as compared with the less salient stimulus and response.

In Study 1, nausea was induced in 50 healthy volunteers through motion sickness in a rotating chair. Subjects were divided into two groups and drank specific beverages prior to the rotation: One group received a different novel-tasting beverage (eg, elderberry, sloe) each day, and the other group drank only water. Nausea-related symptoms were recorded before and after the rotation. Results showed that subjects who drank the novel-tasting beverages reported less AN than those who drank water.

Study 2 tested the protocol from Study 1 in cancer patients undergoing chemotherapy. Immediately prior to chemotherapy infusion, one group received a different novel-tasting beverage each day and the other group drank only water. Results again demonstrated that subjects in the group with different novel-tasting beverages experienced less AN than those who drank water.

Study 3 researchers investigated whether patients' pretreatment expectations of developing nausea or medication-induced post-chemotherapy nausea (PN) were a better predictor of subsequent AN. Before their first round of chemotherapy (session #1), subjects were surveyed about their degree of certainty of developing nausea. PN was measured after session #1, and then AN was measured before session #3 (Figure 1).

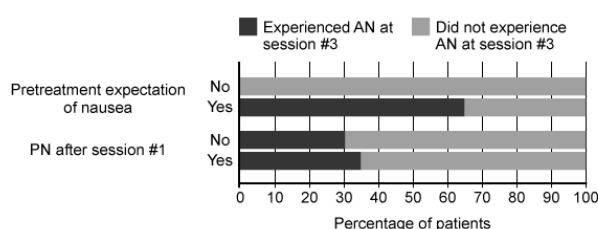


Figure 1 AN before session #3

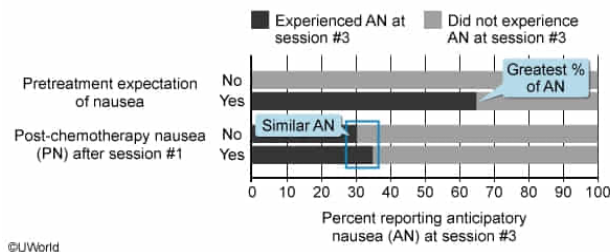
Based on Figure 1, which conclusion is most likely to be supported?

- ✔ ☒ A. Pretreatment expectation of nausea is correlated with AN at session #3. [73%]
- ☐ B. Pretreatment expectation of nausea is correlated with PN at session #1. [10%]
- ☐ C. Experiencing PN at session #1 is correlated with AN at session #3. [12%]
- ☐ D. Lack of PN at session #1 is correlated with AN at session #3. [4%]

Correct

73% answered correctly

Explanation:



When random assignment to groups is not possible (or ethical) in research involving human subjects, researchers will often conduct a **correlational research** study. In these designs, **predictor variables** may be assessed for their association with **outcome variables**, although specific causal relationships cannot be determined.

In Study 3, researchers investigated whether a pretreatment expectation of nausea or experience of PN from session #1 was a better predictor of AN at session #3. Because they could not randomly assign subjects to the condition of whether or not to expect nausea, this is a correlational study design.

Stacked bar graphs are useful for depicting and comparing parts of a whole, with different colors representing different **subgroups** within a **larger group**. Figure 1 illustrates that the group with pretreatment expectations of nausea contained the highest percentage of those who went on to develop AN, which supports the role of cognitive processes (eg, expectations) in associative learning.

(Choice B) Pretreatment expectation of nausea as a predictor of PN at session #1 is not shown in the graph.

(Choices C and D) The percentage of patients who went on to develop AN at session #3 was similar to the percentage of patients who did and did not experience PN at session #1. Neither PN nor a lack of PN at session #1 was as strongly associated with AN at session #3 as pretreatment expectation of nausea.

Educational objective:

When random assignment to conditions is not possible, researchers may use a correlational study design in which predictor variables may be assessed for their association with outcome variables. Correlational designs do not allow for the determination of causal relationships. Stacked bar graphs depict data with subgroups that make up the whole.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:401406

System:
Learning, Memory, and Cognition